

Skills Summary

Comparing Groups with Two-Way Tables

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a cell and a total

How a two-way table is built: one variable labels the rows, the other labels the columns, and each inner cell counts the people who are in *both* of those categories.

The last column holds each row's total, the last row holds each column's total, and the corner holds the grand total. Adding a row across, or a column down, must reproduce those totals.

Example: How many people are in group *B* and answered yes?

	Yes	No	Total
Group A	14	6	20
Group B	9	11	20
Total	23	17	40

Step 1. Two categories at once means one inner cell, so find the row first and then the column.

Step 2. Run along the Group B row to the Yes column; the cell where they meet holds the count. $\Rightarrow$ 9

Try it: In a table, Group A has 21 yes and 9 no, and Group B has 13 yes and 17 no. How many are in group A and answered no?

check: 6

Skill 2 — Turning a count into a row percentage

Rule: $\text{row percentage} = \frac{\text{cell}}{\text{that row's total}} \times 100$

The denominator is the total of the row the cell sits in — never the grand total, unless the question really is about everybody. Choosing the wrong denominator is the most common mistake with these tables.

Example: A row of 30 people contains 12 who said yes. What percent of that row said yes?

Step 1. The question is about that row only, so the row total is the denominator.

Step 2. $\frac{12}{30} \times 100 = 0.4 \times 100 \Rightarrow$ 40

Try it: A row of 24 people contains 18 who said yes. What percent of that row said yes?

check: 75

Skill 3 — Comparing two groups fairly

Rule: compare *fractions of each row*, not the raw counts.

Two rows almost never hold the same number of people, so a bigger count can come from a bigger group rather than a stronger tendency. Write each group's share as a fraction of its own row total, simplify, then compare.

Example: Group A: 12 out of 30 said yes. Group B: 18 out of 24 said yes. Which share is larger?

Step 1. The row totals differ, so the counts cannot be compared directly — build a fraction for each row.

Step 2. $\frac{12}{30} = \frac{2}{5} = 0.4$ and $\frac{18}{24} = \frac{3}{4} = 0.75$, so group B has the larger share. $\Rightarrow \frac{2}{5} < \frac{3}{4}$

Try it: Group A: 21 out of 30. Group B: 26 out of 40. Which share is larger?

check: $\frac{21}{30} < \frac{26}{40}$

Skill 4 — Deciding whether there is an association

Rule: work out each row's percentage for the *same* column and look at the gap.

A large gap means the two variables appear **associated** — knowing the row tells you something about the answer. A gap of a percentage point or two means little. An association is not proof that one thing causes the other.

Example: Row one: 42 of 50 passed. Row two: 30 of 50 passed. How many percentage points separate them?

Step 1. Association is judged from the gap between row percentages, so compute both and subtract.

Step 2. $\frac{42}{50} \times 100 = 84$ and $\frac{30}{50} \times 100 = 60$, so the gap is $84 - 60 \Rightarrow$ **24** percentage points — a large gap, so the variables appear associated

Try it: Row one: 45 of 60 liked it. Row two: 24 of 60 liked it. How many percentage points separate them?

check: 25